1. You designed a new floating-point unit that executes floating-points instructions 5 times faster. The execution time of a program with old floating-point unit is 100 seconds and the execution time with the new floating-point unit is 40 seconds. What portion of the original execution time (of 100 seconds on the old hardware) are floating point instructions?

2. Consider a machine which showed the following benchmark results

Instruction Type	Cycles to execute	Frequency
A	1	50%
B	2	25%
C	3	15%
D	4	10%

What is the CPI? Assuming a 1GHz clock, what is the MIPS rating?

3. Write ARMv8 (or LEGv8) code for the following C segment. Assume X0 contains the base address of A.

```
if (A[5] > 0) A[5] = A[4] +1;
else A[5] = A[6]+1;
```

4. Translate the following Assembly language code into C. Assume X10 contains loop index i, X0 holds *result* and X1 contains the base address of an array A.

```
        SUB   X10, X10, X10
        ADD   X0, X10, #1
LOOP:   LDUR X11, [X1, #0]
        MUL   X0, X0, X11
        ADD   X1, X1, #8
        ADD   X10, X10, #1
        SUBS  XZR X10, #100
        B.LT   LOOP
Exit:
```

5. Consider a machine which showed the following benchmark results

Instruction Type	Cycles to execute	Frequency
A	1	30%
B	2	25%
C	3	20%
D	4	15%
E	5	10%

What is the CPI for machine based on this data? Assuming a 125Mhz clock, what is the MIPS rating?

The compiler group decided to improve the performance. The new optimization resulted in the following.

Instruction type	Percentage of instructions as compared to unoptimized programs
A	95%
B	85%
C	80%
D	90%
E	75%

That is, if the unoptimized program executed 200 type B instructions, the optimized program will need only 170 type B instructions.

What is the new CPI after the compiler optimization? What is the MIPS rating?

6 Consider the base machine of problem 5 above. The hardware designers felt that they can optimize the implementation of their instructions. They also feel that the clock rate can be increased to 150 Mhz. The following is the data on the new machine.

Instruction Type	Cycles to execute	Frequency
A	1	35%
B	2	20%
C	3	15%
D	4	15%
E	5	10%
F	7	5%

Compute the new CPI and MIPS rating and compare the performance improvement as a result of the hardware enhancements.

7 Implement the following conditions in ARMv8 (or LEGv8) assembly langauge

(a) if ($i \geq j$) (b) if ($i \leq j$) (c) if ($i > j$)

Assume the values of i and j are in X1 and X2

8 Write a LEGv8 assembly code to compute the sum of five values. The values are stored in registers X10, X11, X12, X13, and X14. Store the sum in register X20. Assume all values are 64-bits. Comment your assembly code.